Team Application Exercises (tAPP-7)

`Instructions:` Work on `problem 1` on your own for 5 minutes. Then discuss your code with your team members and present one solution as a team (5 minutes). Follow the same process for all the tasks. Swap your solutions with another team for peer evaluation (15 minutes).

Problem 1: The following method is intended to find and return the minimum value from a list of numbers. Identify the bug in the method and indicate how it could be fixed.

```
public int findMinValue(int[] ints) {
    int min = 0;
    for (int i : ints) {
        if (i < min) {
            min = i;
        }
    }
    return min;
}
```

Problem 2:

Consider the following Java method:

```
public static void main(String[] args) {
    System.out.println("Hello world");
}
```

Answer the following questions in the context of the above method signature:

1. What is special about a method with the given signature?
2. What effect does the **public** keyword have on this method? What would be the difference if it were removed?
3. What effect does the **static** keyword have on this method? What would be the difference if it were removed?
4. What is the meaning of **void** in the method signature? What would be the effect if it were removed?
5. What data type is **String[] args**? What is another way that same data type could have been written?

Problem 3: Fill in the most appropriate Java code into the blank spaces in the following list of class declarations. (There may be different possible answers for some blanks – this may provoke useful discussions within your team!)

// file animals/Animal.java



```
public abstract class Animal {

      [redacted] String name;

      public Animal(String name) {
            [redacted]
      }
      public [redacted] void move();
}
```

// file dogs/Dog.java



```
public class [redacted] Animal {

      public Dog(String name) {
            [redacted]
      }

      [redacted] move() {
            System.out.println("Dogs can run");
      }

}
```